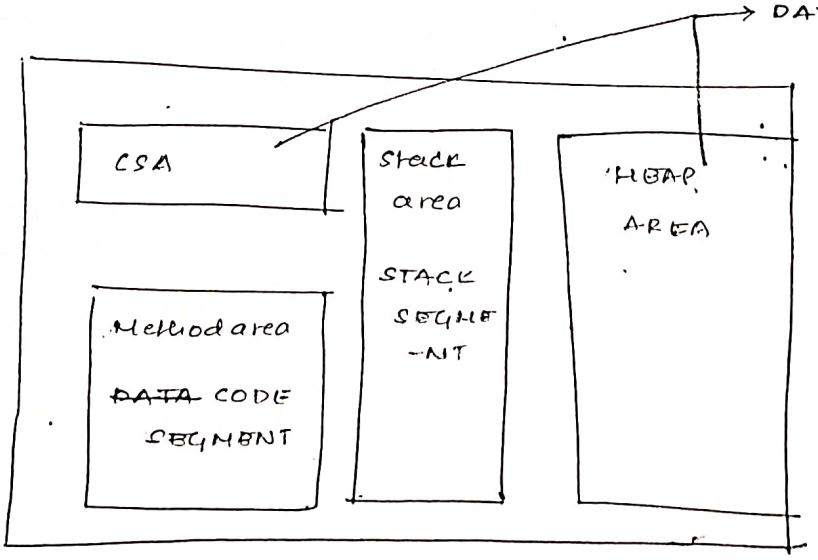
THREADS

Smallest unit or sequence of execution in a process i
known as Thread.

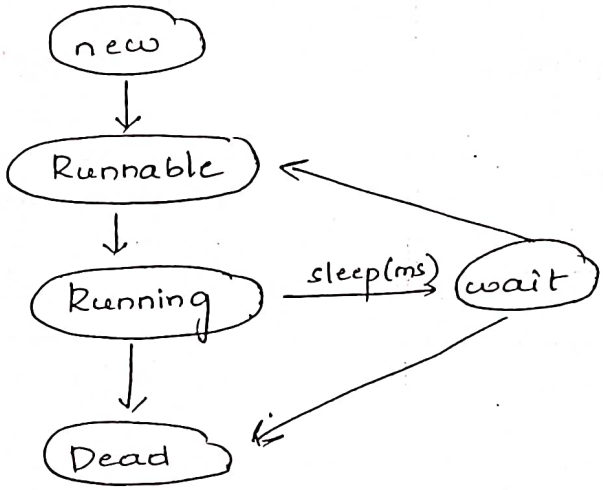
NOTE:

1. Every thread will have its own stack and Program counter
2. One process can have multiple threads, it is known as Multiple Threading.
3. Multiple Threading is required to achieve parallel execution (
that is to execute more than one instruction at a same time).
4. All the threads of the process share the resource of th
Process. ie all the threads share the same code segment of
the process. & same data segment. But they will have differ
stack stagment



THREAD LIFE CYCLE:

NEW STAGE	When a thread is just created, it is in new stage. It remains to stay in same stage until it is started.
RUNNABLE STAGE	When a new thread is started, the thread goes to runnable stage. In this stage the thread is considered to be ready for execution.
RUNNING STAGE	The thread which is under execution is considered to be in running stage.
WAIT STAGE/ BLOCK STAGE	When the thread ~~be~~ in running stage is stopped, the thread is in wait stage. It remains to be in the same stage until: 1. Another thread signals to the thread which is in waiting stage. 2. Timed wait: It stays in a wait stage for a specified time, after which the thread goes back to runnable stage.
DEAD STAGE/ TERMINATE STAGE	A thread goes to dead stage once the given task completed. (If any thread reaches this stage, the stac created will be removed).



java.lang.Thread :

Thread is a built-in class which is present inside lang package. It helps the programmer to have their own threads created by extending the classes to thread class.

The thread class extends to the object class & implements Runnable interface.

In java, we have few default threads like main, scheduler etc & java also provides authority to the user to create his own threads ie user defined threads.

Any thread created will have three important states as follows:

1) Thread id
2) Thread Name
3) Thread priority.

To get the details of states we need the address of current thread object which can be obtained with the help of a static method present inside Thread class called as currentThreads() method.

METHODS TO MOVE THREADS FROM ONE STAGE TO ANOTHER STAGE:

1) start():

It moves the thread from new stage to Runnable stage It has to be called explicitly.

2) run():

It is used to move the thread from Runnable stage to running stage. It will be called implicitly whenever we start a thread.

3) sleep(long ms):-

It is used to move the Thread from Running stage to wait stage for specified time which the user will pass during method call. Sleep is a static method which is present inside thread class & it throws an exception called as InterruptedException

Therefore it becomes mandatory for the programmers either to handle the exception or to throw the excepti

whenever they call sleep()

USER DEFINED THREADS:

A user can also create his own thread by following ways

STEP 1 : Create a class.

STEP 2 : 1. Either ~~inherit~~ extend the Thread class

or

2. Implement the Runnable interface.

Ex: class Employee extends Thread

```
{
 String name;
 Employee() { }
 Employee (String name)
 {
 this.name= name;
 }
 public void print()
 {
  for (int i=0; i<5; i++)
  {
    System.out.println ("Name : "+ name + " : " + i);
    try
    {
      Thread.sleep (2000);
    }
    catch (InterruptedException e)
    {
    System.out.println ("Exception Handled");
    }
  }
}
public void run()
{
  print();
}
```

```
class Driver1
{
  public static void main (String [] args) throws Exception
  {
    Employee e1 = new Employee ("Smith");
    Employee e2 = new Employee ("Adams");
    Employee e3 = new Employee ("miller");

    e1.start();
    e2.start();
    e3.start();
  }
}
```